

Global Sorting, Morse Indices, and the Route-A Boundary of the Brockett Double-Bracket Flow

Route-A structural certificate C185

Abstract

For every real symmetric simple-spectrum orthogonal orbit and every strictly increasing diagonal target, we close the Brockett flow $\dot{H} = [H, [H, N]]$ without a dimension cutoff. The flow is globally isospectral and satisfies $d\text{Tr}(HN)/dt = \|[H, N]\|_F^2$. Its $n!$ equilibria are diagonal permutations. Each tangent pair has an explicit rate whose positive signs are precisely the inversions; hence the Morse index of the sorting energy is the inversion number. Every trajectory converges, generic data sort, and no nonconstant trajectory is recurrent.

Keywords: double bracket; isospectral flow; matrix sorting; Morse index; gradient dynamics.

中文摘要

本文对任意维实对称单谱正交轨道与严格递增对角目标，完整刻画 $\dot{H} = [H, [H, N]]$ ：解全局存在并保持谱，势函数导数恰为交换子的 Frobenius 范数平方；全部平衡点是 $n!$ 个对角排列。每个切向二元模态的符号由排列逆序决定，从而排序能量的 Morse 指标等于逆序数；所有轨道收敛，几乎所有初值完成排序，且不存在非常值回复或周期轨道。
关键词：双括号流；等谱流；矩阵排序；Morse 指标；梯度动力学。

1 Frozen family and global identities

Let

$$\lambda_1 < \dots < \lambda_n, \quad \nu_1 < \dots < \nu_n,$$

put $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ and $N = \text{diag}(\nu_1, \dots, \nu_n)$, and work on $\mathcal{O}_\Lambda = \{Q\Lambda Q^\top : Q \in O(n)\}$. The equation and its gradient sorting role are classical Brockett [1]; no priority is claimed here.

Set $K(H) = [H, N]$. It is skew-symmetric and $\dot{H} = [H, K(H)]$. If $\dot{Q} = -K(H)Q$, orthogonality is preserved and $H = Q\Lambda Q^\top$ solves the equation. Thus the compact orbit is invariant, all spectral traces are constant, and the smooth vector field is complete. Cyclicity of trace gives the strict identity

$$\frac{d}{dt} \text{Tr}(HN) = \text{Tr}([H, K]N) = -\text{Tr}(K^2) = \|K\|_F^2. \quad (1)$$

Consequently equilibrium is equivalent to $[H, N] = 0$. The strict target makes H diagonal, and the simple orbit spectrum makes its diagonal a permutation of the λ_i . There are exactly $n!$ equilibria $D_\pi = \text{diag}(\lambda_{\pi(1)}, \dots, \lambda_{\pi(n)})$.

Structure	Exact consequence	Boundary
skew Lax generator	global isospectral flow	not a fixed quantum operator
squared-norm derivative	strict gradient ordering	no nonconstant recurrence
simple/strict spectra	$n!$ isolated equilibria	target repetitions give zero modes

2 Pair modes, Morse index, and convergence

At $D = D_\pi$, write $H = D + \varepsilon X$. Since $[D, N] = 0$, linearization gives

$$L_D X = [D, [X, N]], \quad (L_D X)_{ij} = (\lambda_{\pi(i)} - \lambda_{\pi(j)})(\nu_j - \nu_i)X_{ij}. \quad (2)$$

The tangent space consists of arbitrary symmetric off-diagonal pair modes. For $i < j$, the second factor in (2) is positive, while the first is positive exactly when $\pi(i) > \pi(j)$. Hence the ascent flow has $\text{inv}(\pi)$ unstable modes and $\binom{n}{2} - \text{inv}(\pi)$ stable modes. Equivalently, the Morse index of $-\text{Tr}(HN)$ is $\text{inv}(\pi)$. This sign convention is essential: it is not the Morse index of the increasing height.

The rearrangement inequality makes increasing alignment the unique height maximum and the only stable equilibrium. The bounded height has a limit; LaSalle invariance places every omega-limit set among the finite equilibria. Connectedness of an omega-limit set then forces convergence to a single one. All equilibria are hyperbolic by (2). Stable manifolds of nonsorted equilibria have lower dimension, so their finite union has zero orbit volume; every other initial point converges to the sorted diagonal. Finally, (1) is strict on a nonconstant trajectory. A recurrent return would restore an old height value, which is impossible; periodicity is a special case.

Revision-round focus. Round 1 adds every pair-mode sign, the inversion/Morse convention, compact convergence, generic sorting, and the recurrence obstruction.

Limitations and nonclaims. The paper does not claim novelty for the Brockett flow, a full degenerate critical-set or cell-closure theorem, arithmetic local data, an Euler factor, a root number, automorphy, a target divisor, a Hilbert–Polya operator, or Route-B authorization. The single source is an ownership lock, not an exhaustive literature review. Internal drafting checks are not external peer review.

Declarations

Data and code availability. Package-local exact evidence and deterministic code accompany this manuscript.

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AI-use disclosure. An AI coding assistant supported drafting and exact-code development; it was not an external reviewer or independent error process.

References

- [1] R. W. Brockett, “Dynamical systems that sort lists, diagonalize matrices, and solve linear programming problems,” *Linear Algebra and its Applications* 146 (1991), 79–91. DOI: 10.1016/0024-3795(91)90021-N.